# Amirhossein Kazerouni

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## Research Summary

PhD researcher working on generative models and 3D/4D reconstruction. Current work includes a geometry foundation model for panoramic video with feed-forward Gaussian reconstruction (Samsung AI Center Toronto) and 4D face generation controllable through facial action units. First- and co-first-author papers at NeurIPS, CVPR, ICCV, and WACV; Samsung Best Paper Award (Bronze) for Face2Scene.

## Education

**University of Toronto** — Toronto, Canada
*Ph.D. in Computer Science, advised by Babak Taati and Michael Brudno* — *Jan. 2024 – Present*

**Iran University of Science and Technology (IUST)** — Tehran, Iran
*B.Sc. in Electrical Engineering (Control)* — *Sep. 2017 – Feb. 2022*

## Research Experience

**Samsung AI Center Toronto** — Toronto, Canada
*Research Intern. Supervisors: Alex Levinshtein, Iqbal Mohomed, Konstantinos G. Derpanis* — *May 2025 – Present*

- **Panoramic 4D Reconstruction:** Developing a geometry foundation model for panoramic video, and a compact feed-forward Gaussian reconstruction model built on its estimated geometry.
- **Face-Guided Scene Restoration:** Estimates the degradation from a low-/high-quality face pair, with the high-quality face from reference-based restoration, to guide one-step diffusion restoration of the scene. *CVPR 2026.*
- **VLM Test-Time Scaling:** Developed a deployment-friendly policy that prunes redundant visual tokens and spends the saved compute on more self-consistency rollouts, improving accuracy at matched cost. *NeurIPS 2026.*
- Contributed to BurstGP, raw burst image super-resolution with generative priors. *arXiv 2026.*

**University of Toronto & Vector Institute** — Toronto, Canada
*Graduate Research Assistant. Supervisors: Babak Taati, Michael Brudno* — *Jan. 2024 – Present*

- **Controllable 4D Generation:** Using a video diffusion model to estimate a person's 3D face in a neutral expression from a single image, then controlling its expressions with facial action units.
- **Data-Agnostic Encoding:** Developed a meta-learned encoder for images and 3D shapes using parallel local implicit functions and hierarchical latents, for generative modeling and classification at lower compute. *ICCV 2025.*
- **Unified Saliency Prediction:** Developed a single Mamba-based saliency model that adapts to different image types instead of one model per domain, to better model human visual attention. *WACV 2025, oral.*

**RWTH Aachen University** — Aachen, Germany
*Research Assistant. Supervisors: Dorit Merhof, Reza Azad* — *2022 – 2024*

- **Medical Image Segmentation:** Built 2D/3D CNN–Transformer and diffusion models with multi-scale fusion, frequency attention, self-supervision, and boundary-aware training. *WACV, MICCAI, ISBI, MIDL (2 orals).*
- **Prior-Conditioned Implicit Representations:** Developed a harmonizer network that uses deep prior knowledge to tune the sinusoidal activations, improving signal fitting, denoising, and super-resolution. *WACV 2024.*

**DGSculptor** — Montreal, Canada
*Machine Learning and Computer Vision Researcher. Supervisor: Alireza Morsali* — *2021 – 2023*

- **Ensemble Neural Representations:** Co-developed an ensemble implicit neural representation in which independent tiny MLPs each fit part of the signal, reaching higher PSNR than SIREN with fewer FLOPs.

**IUST AI and Control Lab** — Tehran, Iran
*Computer Vision Researcher (undergraduate)* — *2019 – 2021*

- Developed a fusion-based multi-stage method for 3D shape analysis of vehicles with a stereo camera, used to measure load protrusion.

**University of Tehran** Tehran, Iran
*Computer Vision Researcher. Supervisor: Mohammad Ali Akhaee* *2019 – 2020*

- **Football Analysis:** Built a bird's-eye-view module: a GAN extracts field lines from broadcast frames, and the homography estimated from them maps each frame onto a top-down pitch template. [Project] [Code]

## Selected Publications

* equal contribution. Full list on Google Scholar.

- **Face2Scene: Using Facial Degradation as an Oracle for Diffusion-Based Scene Restoration**
  **Amirhossein Kazerouni**, Maitreya Suin, Tristan Ty Aumentado-Armstrong, Sina Honari, Amanpreet Walia, Iqbal Mohomed, Konstantinos G. Derpanis, Babak Taati, Alex Levinshtein. *CVPR 2026* **SBPA Bronze Award** [Project]
- **AVIS: Adaptive Test-Time Scaling for Vision-Language Models**
  Ahmadreza Jeddi*, Minh Ngoc Le*, **Amirhossein Kazerouni***, Hakki Can Karaimer, Hue Nguyen, Iqbal Mohomed, Michael Brudno, Alex Levinshtein, Konstantinos G. Derpanis, Babak Taati, Radek Grzeszczuk. *NeurIPS 2026* [Project]
- **LIFT: Latent Implicit Functions for Task- and Data-Agnostic Encoding**
  **Amirhossein Kazerouni**, Soroush Mehraban, Michael Brudno, Babak Taati. *ICCV 2025* [Project]
- **SUM: Saliency Unification through Mamba for Visual Attention Modeling**
  Alireza Hosseini*, **Amirhossein Kazerouni***, Saeed Akhavan, Michael Brudno, Babak Taati. *WACV 2025* **Oral** [Project]
- **INCODE: Implicit Neural Conditioning with Prior Knowledge Embeddings**
  **Amirhossein Kazerouni**, Reza Azad, Alireza Hosseini, Dorit Merhof, Ulas Bagci. *WACV 2024* [Project]
- **PriLoRA: Prior-Conditioned Low-Rank Adapters for Medical Vision Models**
  **Amirhossein Kazerouni***, Sanaz Karimijafarbigloo*, Ali Aghayari*, Ali Ghasemzadeh Khorzoughi*, Reza Azad, Wojciech Samek, Dorit Merhof, Michael Brudno, Babak Taati. *MICCAI 2026* [Code]
- **HiFormer: Hierarchical Multi-Scale Representations Using Transformers for Medical Image Segmentation**
  Moein Heidari*, **Amirhossein Kazerouni***, Milad Soltany*, Reza Azad, Ehsan Khodapanah Aghdam, Julien Cohen-Adad, Dorit Merhof. *WACV 2023* [Code]
- **OpenVAM: Open-World Visual Attention Modeling with VLMs**
  Kiana Hooshanfar*, **Amirhossein Kazerouni***, Alireza Hosseini*, Michael Brudno, Babak Taati. *WACV 2027* [Project]
- **A Unified Theory of Sinusoidal Activation Families for Implicit Neural Representations**
  Alireza Morsali, MohammadJavad Vaez, Mohammad Hossein Soltani, **Amirhossein Kazerouni**, et al. *Transactions on Machine Learning Research (TMLR), 2026* [Project]

## Honors and Awards

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| 2026 | **SBPA Bronze Award** (Samsung Best Paper Awards, Samsung Group-wide), for Face2Scene. |
| 2026 | **Outstanding Reviewer**, ECCV 2026. |
| 2024–2026 | **Vector Research Grant**, Vector Institute. |
| 2021 | **1st place (team)**, national Rahneshan autonomous vehicle competition, INEF, Iran. |
| 2021 | **2nd place (team)**, FIRA World Cup, Autonomous Cars League. |

## Academic Service

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| Conference reviewer | CVPR 2026, ECCV 2026 (Outstanding Reviewer), NeurIPS 2024–2026, AAAI 2027, MICCAI 2024, 2026 |
| Journal reviewer | Medical Image Analysis; Engineering Applications of Artificial Intelligence |

## Teaching and Mentoring

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| Teaching Assistant | University of Toronto: CSC2503 Computational Vision; CSC420 Image Understanding (3 terms); CSC384 Artificial Intelligence; CSC148 Computer Science |
| Co-founder | AIR (Artificial Intelligence and Robotics) Center, Tehran (2020–2022), a training center offering courses in deep learning, machine learning, and Python |

## Skills

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| Programming | Python, C/C++, MATLAB; PyTorch, JAX, TensorFlow, OpenCV, NumPy; Linux, Git |